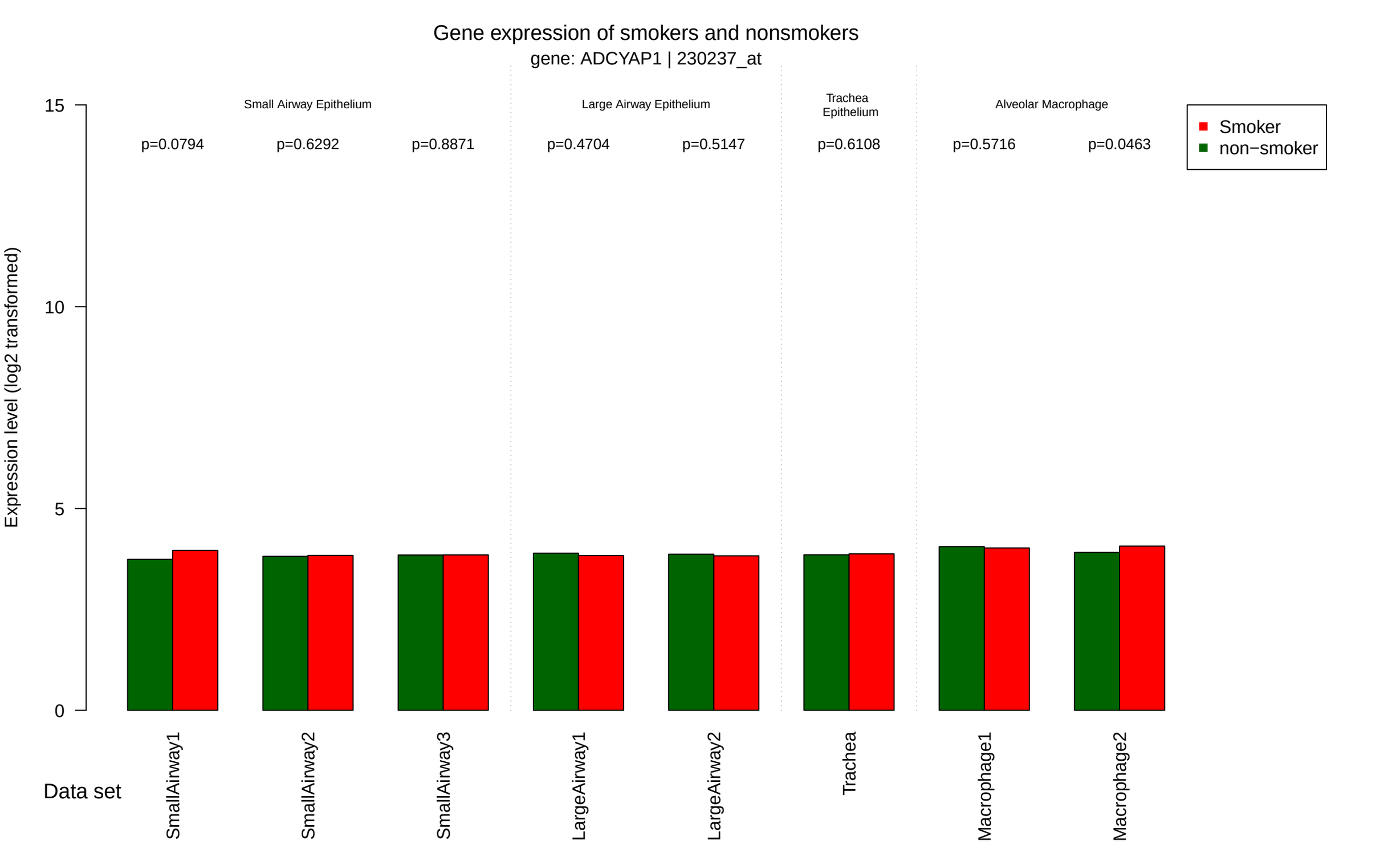
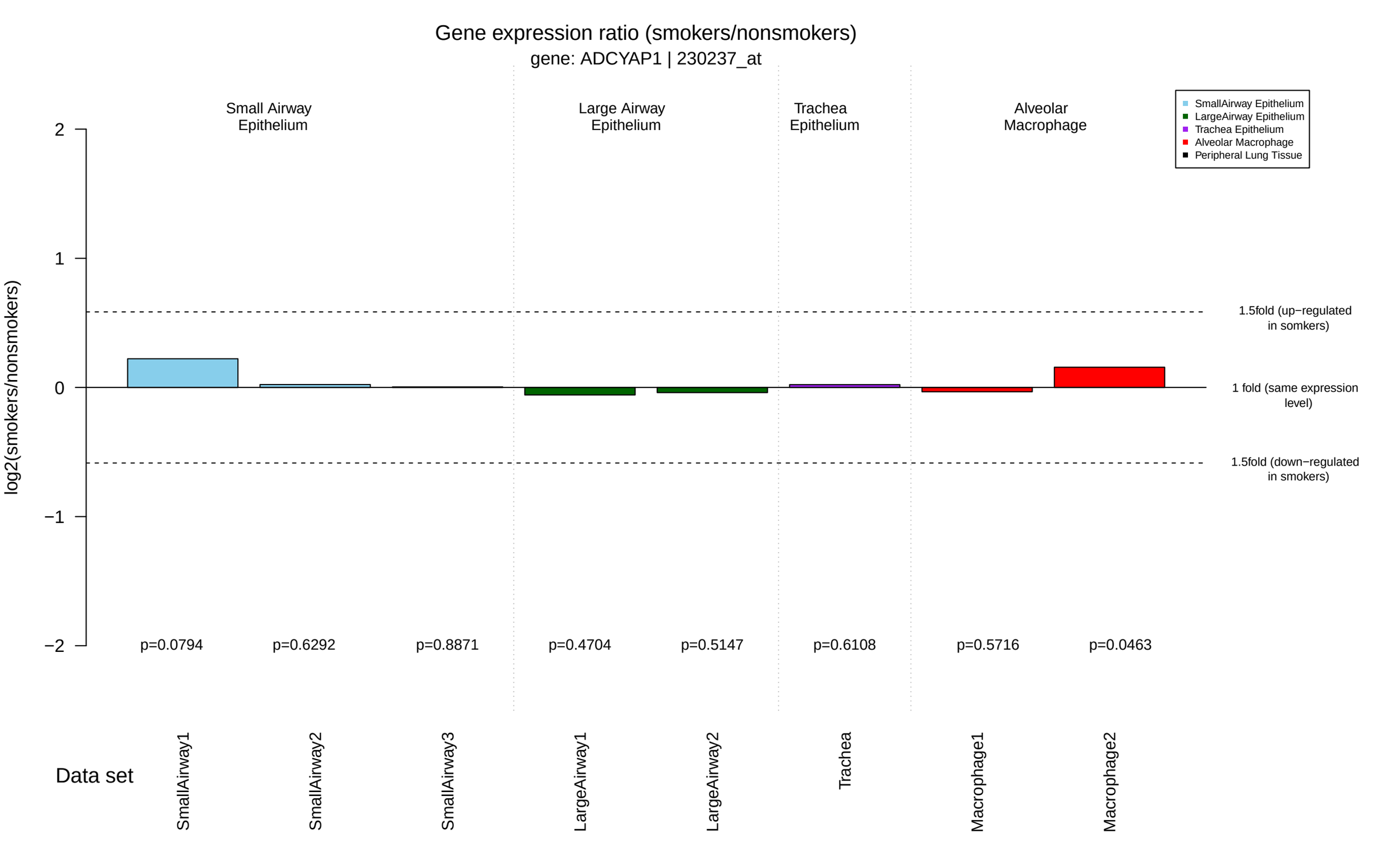
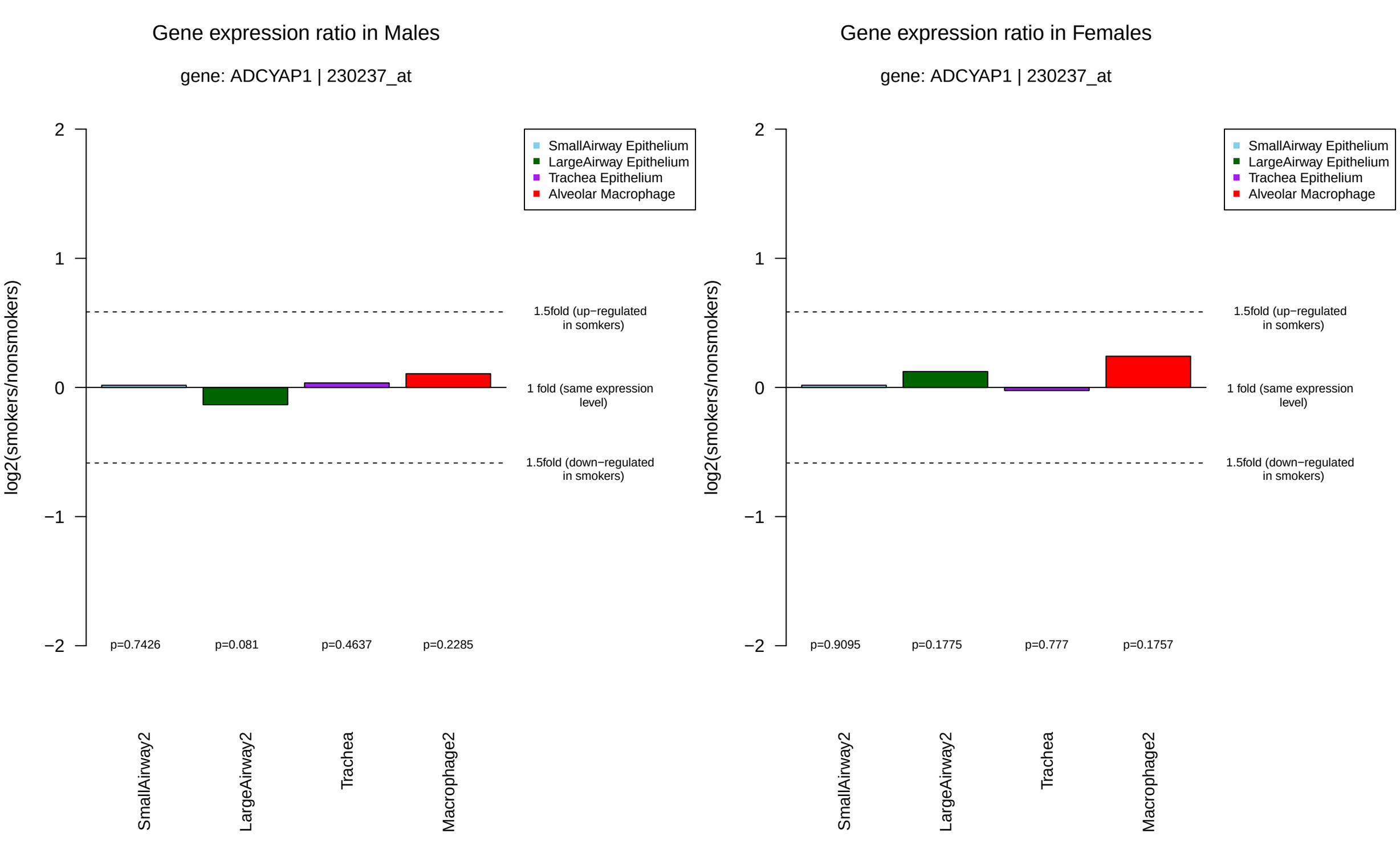
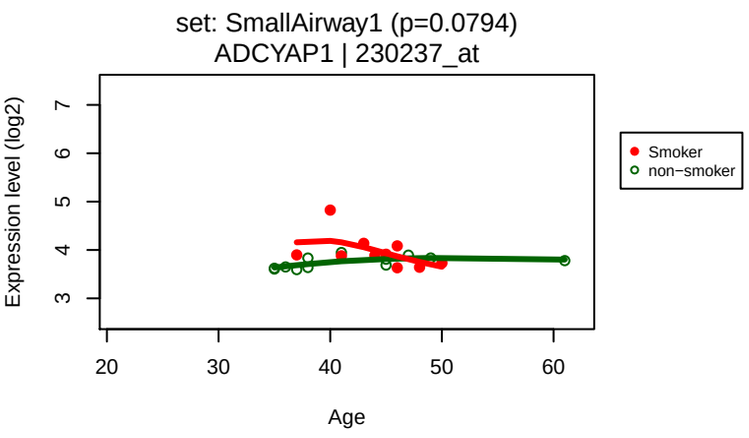
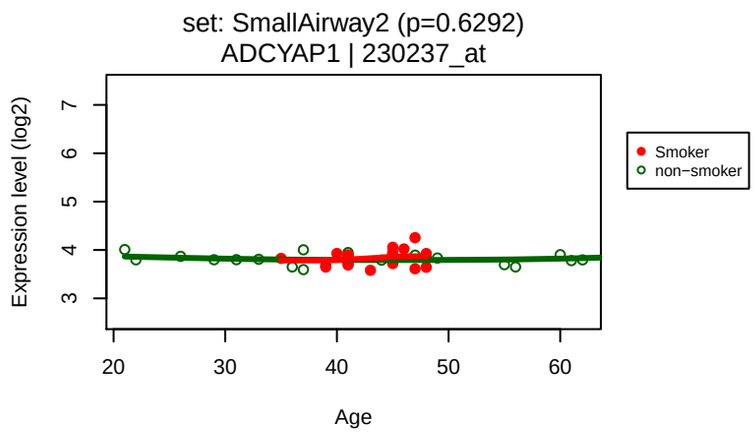
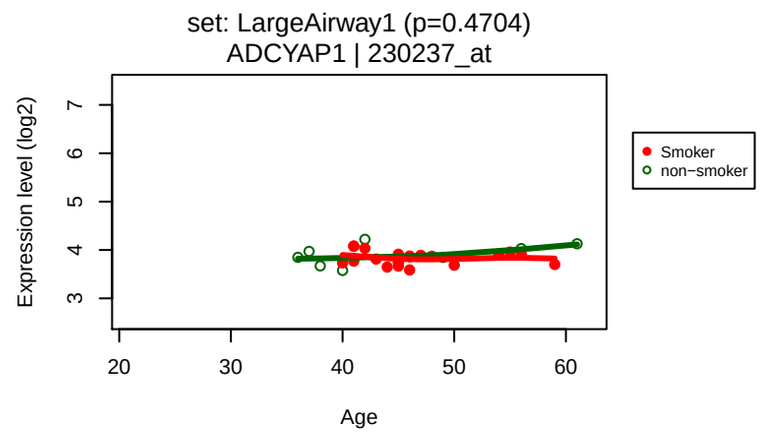

Gene expression of smokers and nonsmokers
gene: ADCYAP1 | 230237_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0794
p=0.6292
p=0.8871
p=0.4704
p=0.5147
p=0.6108
p=0.5716
p=0.0463
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

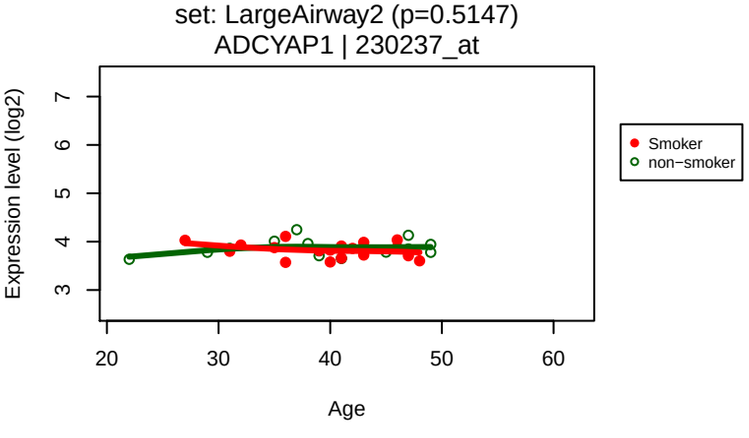

Gene expression ratio (smokers/nonsmokers)
gene: ADCYAP1 | 230237_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0794
p=0.6292
p=0.8871
p=0.4704
p=0.5147
p=0.6108
p=0.5716
p=0.0463
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

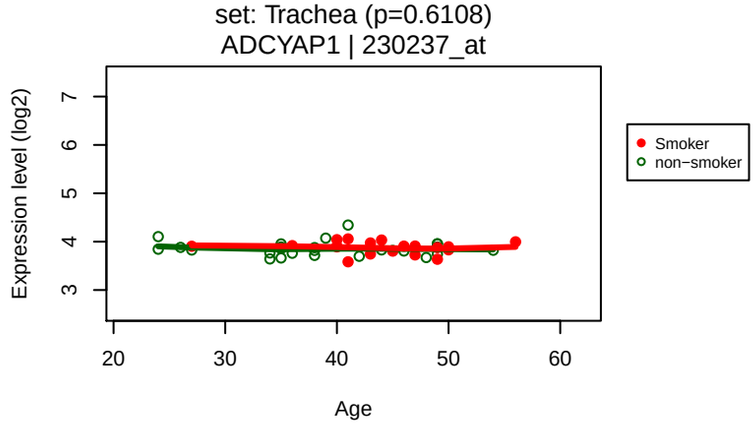

Gene expression ratio in Males
gene: ADCYAP1 | 230237_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7426
p=0.081
p=0.4637
p=0.2285
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADCYAP1 | 230237_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9095
p=0.1775
p=0.777
p=0.1757
SmallAirway2
LargeAirway2
Trachea
Macrophage2

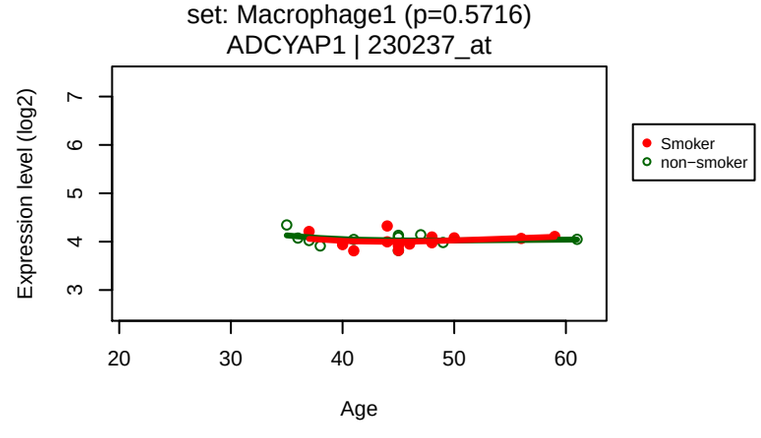

set: SmallAirway1 (p=0.0794)
ADCYAP1 | 230237_at
Expression level (log2)
Age
Smoker
non−smoker

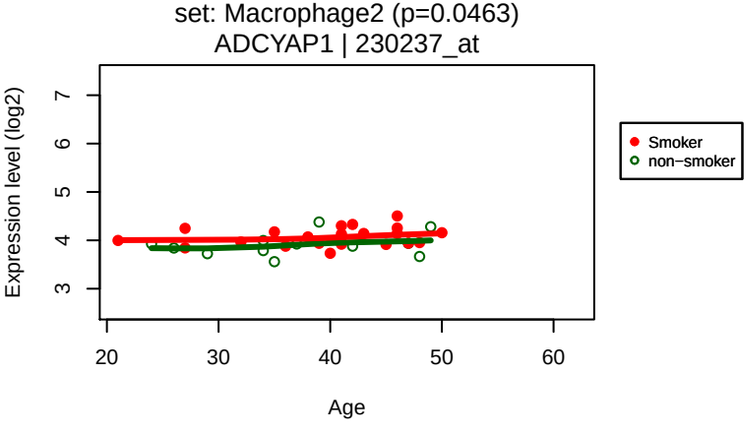

set: SmallAirway2 (p=0.6292)
ADCYAP1 | 230237_at
Expression level (log2)
Age
Smoker
non−smoker

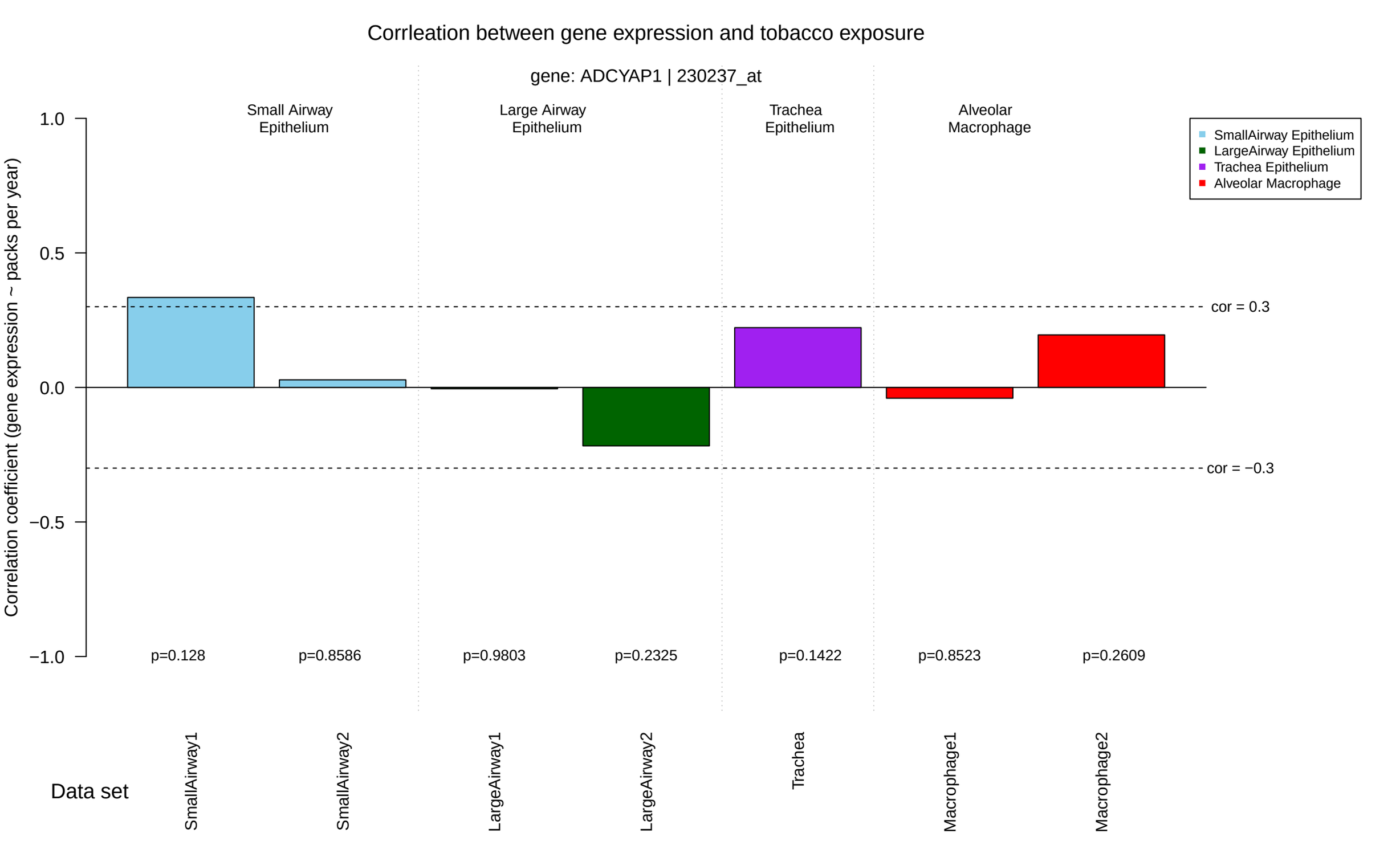

set: LargeAirway1 (p=0.4704)
ADCYAP1 | 230237_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5147)
ADCYAP1 | 230237_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6108)
ADCYAP1 | 230237_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5716)
ADCYAP1 | 230237_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0463)
ADCYAP1 | 230237_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADCYAP1 | 230237_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.128
p=0.8586
p=0.9803
p=0.2325
p=0.1422
p=0.8523
p=0.2609
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2